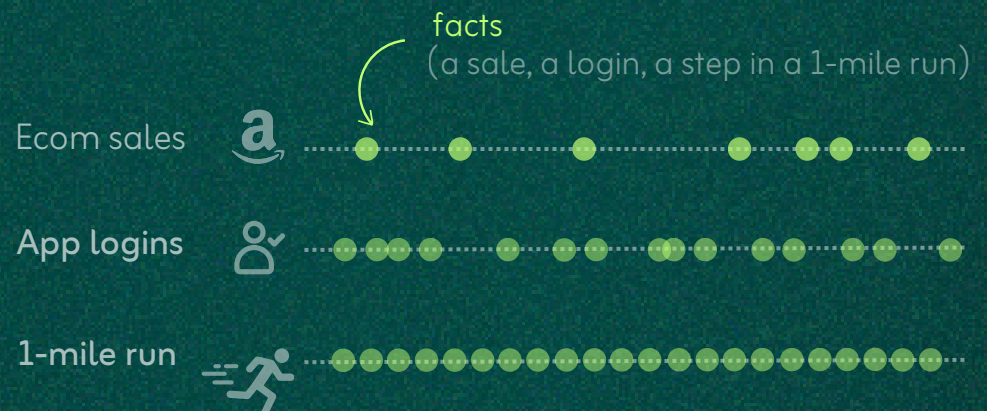


1 FACT ~ Truth

- Atomic (can't be broken down into smaller units)
- Can be aggregated up/down
- Not slowly changing
- Can't be changed back



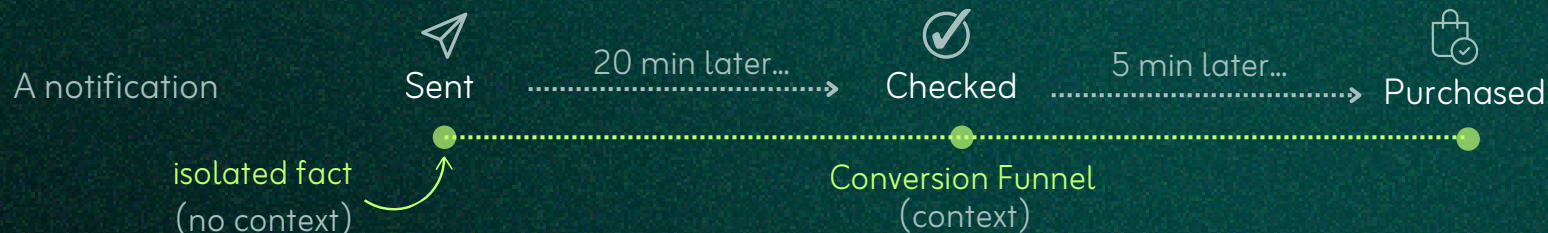
? Why Fact Data is hard to model?

1. Scalability (10-100x the amount of dimension data)

Q: How many actions taken per dimension?

ie. : 2 bio users x 30 notifications/day ~ 50 bio notifications/day

2. Need more context for effective analysis



Q: What data dimensions do we need to make the fact data more valuable?

ie. A user's location bring context (what's the clickthrough rate between US and Canada notification recipients?)

3. Duplicates are more common. Causes:

- Bugs
ie. A software engineer pushed a bug in the logger> 1 click generates 2 logs> Data Quality issue
- Genuine duplicates (can be handled via deduplication)
ie. A notification clicked now and clicked again 3 hours later
Tracking 2 genuine actions without counting them as 2?> deduplication
No deduplication> your metric will look weird! (ie. a 200% CTR)

? How Fact Data Modeling works?

Normalization

- Splits data across multiple tables
- Minimizes data redundancy
- Removes duplicates, increases data integrity

Denormalization

- Combines data into a single table
- Faster querying (use GROUP BY, not JOIN)
- High storage cost, potential duplications

Trade-off



? Fact Data vs. Raw Logs: same thing?

- No, but both interconnected (no fact without raw logs)
- The level of trust of fact data should be higher than the trust in raw logs

Raw Logs

- Data for systems, server running.
- Usually owned by SEs (software engineers)
- Ugly schemas, potential duplicates, DQ issues
- Short retention

Fact Data

- Data for analysis.
- Usually owned by DEs (data engineers)
- Nice schemas/column names
- Quality guarantess (uniqueness, not nulls, ...)
- Longer retention

💡 Goal as a DE: collaborate closely with SEs, be able to convert raw logs into highly trusted fact data

? Aspects of Fact Modeling

A WHO> 'Who is part of that event?'

- 'Pushed out as IDs (ie. Tesla car> <'user_id', 'car_id', 'operating_system_id'>)

B WHERE> 'Where is the event taking place?'

- Modeled with ids (Who) + additional dimensions
ie. location (Country, City, State); app pages (Homepage, Profile Page... modeled w/ backlashes)

C HOW> 'How is part of the event?'

- Similar to 'Where' and 'Who'
ie. "Make a click using an iPhone"  where (where in the app, which button they clicked)
how (methodology that lead to click)

D WHAT> 'What is the event taking place?'

- Atomic events
ie. 1. Atomic : A notification> 'generated', 'sent', 'clicked', 'converted into purchase'
2. Not Atomic: 'Burning Man'> Higher-level event (group/aggregation of smaller events)

E WHEN> 'When is the event taking place?'

- Modeled as timestamp
- Align all app logins to same time zone (UTC), client-side login (higher fidelity vs server side login)

